

Extension of Myopic Coverage Path Planning from Rover to UAV (Technical Summary & Research Proposal Outline)

This document summarizes the engineering modifications required to adapt the algorithm presented in “Risk-Aware Coverage Path Planning for Lunar Micro-Rovers Leveraging Global and Local Environmental Data” to an Unmanned Aerial Vehicle (UAV). The goal is to maintain the lightweight myopic coverage strategy while extending the system to operate in a 3D exploration environment with aerial vehicle dynamics.

1. Key Structural Changes

Component	Original Rover System	Proposed UAV System
State Space	2D grid (x,y)	3D voxel grid (x,y,z)
Neighbor Set	8 neighboring cells	26 neighboring voxels
Terrain Risk	DEM slope cost	Altitude change + obstacle clearance risk
Energy Model	Wheel rolling energy	Thrust-based flight energy model
Obstacle Avoidance	Bug algorithm	Artificial Potential Field (APF)
SLAM	HDL Graph SLAM	LIO-SAM (LiDAR + IMU)
Map Representation	2D occupancy grid	3D voxel map

2. UAV Mathematical Models

Altitude Risk Cost

$$mc_alt = w1|z(t+1) - z(t)| + w2(1 / d_obs) + w3|z(t+1) - z_ref|$$

Energy Consumption Model

$$E_move = P_hover * t + m*g*(z(t+1) - z(t)) + F_drag * d$$

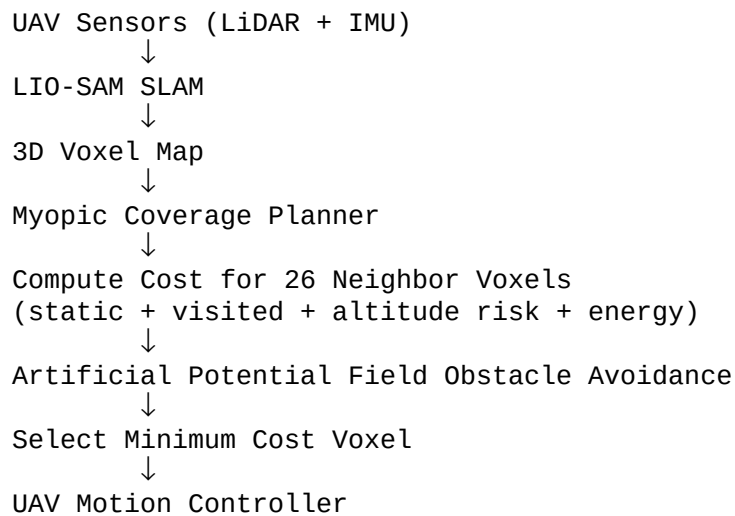
Hover Power Model (Momentum Theory)

$$P_hover = (m*g)^{(3/2)} / \text{sqrt}(2 * \rho * A)$$

Final UAV Cost Function

$$mc = \alpha(mc_static + mc_visited * V_i) + \beta mc_alt + \gamma mc_energy$$

3. Algorithm Flow



4. UAV Coverage Algorithm Pseudocode

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Initialize 3D voxel map
Initialize UAV position

while coverage < target:

    sense nearby voxels
    update occupancy map

    neighbors = generate 26 neighbors

    for each neighbor:
        compute motion cost
        cost = static + visited + altitude + energy

    choose neighbor with minimum cost

    apply APF obstacle avoidance
    move UAV to selected voxel

    update SLAM pose

end
  
```

5. Expected Research Contributions

- Extension of myopic coverage path planning to 3D UAV exploration environments.
- Introduction of altitude-aware risk cost functions for aerial navigation.
- Integration of energy-aware motion planning based on UAV thrust models.
- Development of a lightweight 3D coverage planner compatible with real-time UAV systems.
- Implementation in simulation environments such as MATLAB UAV Toolbox, Gazebo, or

CoppeliaSim.